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Arnoldi-Chebyshev Approach for Convection-Diffusion Computations

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Abstract—In a convection-diffusion equation, if the convection term is very dominant, the linear system of equations which result from either finite-differencing or finite elementing will not have a strictly diagonally dominant coefficient matrix. So, if one tries a conventional iteration method (Jacobi or Gauss-Seidel) to solve the linear system of equations, the iteration matrix may not satisfy the spectral radius condition for convergence, and hence, no converging solution may be obtained. The problem can be overcome, under certain conditions, if one uses a two-step iteration procedure involving the spectral enveloping ellipse for the iteration matrix. In this paper, we present such a two-step method that combines an Arnoldi-Chebyshev approach for convection-diffusion computations.
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Keywords—Convection-diffusion equation, Finite-difference discretization, Iterative methods, Spectrum enveloping ellipse, Arnoldi-Chebyshev approach.

1. INTRODUCTION

If a convection diffusion equation is discretized using the central difference scheme the resulting coefficient matrix will not be diagonally dominant whenever the convection terms are large. So such a system of linear equations solved using a conventional iteration method often fail to converge as some of the eigenvalues of the iteration matrix lie outside the unit circle $C = \{z : |z| \leq 1\}$ in the complex plane.

In a recent article, Gupta [1] used a two-step iterative method developed by de Pillis [2] on a model convection-diffusion problem to overcome the above mentioned difficulty. The two-step method was used in the format of the Jacobi iteration method and converging solutions were obtained even when the convection term was dominant in the model problem. This study was extended in [3] where a Gauss-Seidel method which incorporates the idea of eigenspectrum enveloping ellipse was developed for convection-diffusion computations. It should be noted that under certain conditions, one can also develop two-step iteration methods which do not have to involve spectrum enveloping ellipses (see [4]). This paper follows [3], but differs in the way we choose the ellipse enveloping the eigenspectrum of the iteration matrix. Here, first we find the eigenvalue of the iteration matrix with the largest real part and use that information to construct the ellipse. We employ a hybrid method based on an Arnoldi-Chebyshev approach [5] to achieve this.

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2. TWO-STEP ITERATION METHOD (JACOBI TYPE) ON A MODEL PROBLEM

We consider the convection-diffusion equation

$$\Delta u - R\frac{\partial u}{\partial x} = f(x,y), \qquad (x,y) \in \Omega, \tag{2.1}$$

with

$$U(x,y) = 0, \qquad \text{on } \partial\Omega,$$

where Δ represents the Laplacian operator, $f(x,y)$ is a given function, and R represents the Reynolds number which is a nonzero parameter, as our model problem. When the partial differential equation (2.1) is discretized using the central difference approximation, one obtains the following finite difference equation:

$$U_1 + U_2 + U_3 + U_4 - 4U(x,y) - \frac{Rh}{2}(U_1 - U_3) = h^2 f(x,y)',$$

where U_1, U_2, U_3, and U_4 represent the four neighbouring values of $U(x,y)$ at $(x+h,y)$, $(x,y+h)$, $(x-h,y)$, and $(x,y-h)$, respectively, and h is the mesh width.

In problem (2.1), if the convection term is approximated by the upwind formula

$$\frac{\partial U}{\partial x} \approx \alpha\frac{[U(x+h,y) - U(x,y)]}{h} + (1-\alpha)\frac{[U(x,y) - U(x-h,y)]}{h},$$

one obtains the following finite difference equation:

$$(1 - Rh\alpha)U_1 + U_2 + 1 + (1-\alpha)RhU_3 + (-4 + 2\alpha Rh - Rh)U_4 = h^2 f(x,y),$$

where U_is are as given earlier, and the upwind parameter $\alpha \in [0,1]$.

Once the difference equation at every interior point together with the boundary conditions are assembled, we will have a system of equations of the form

$$AU = b \tag{2.2}$$

to be solved. It can be easily checked that A is strictly diagonally dominant when $|Rh| < 2$, and so, under this condition, if one employs the conventional Jacobi or the Gauss-Seidel iteration method to solve (2.2), convergence can be achieved [6]. However, by making $|R|$ large, for a given mesh width h, $|Rh|$ can be made greater than 2. This in turn will make A lose its strict diagonal dominance and may result in the iteration matrix (for example, Jacobi) having the spectral radius greater than unity. Therefore, using the Jacobi iteration (when $|Rh| > 2$) to solve (2.2) may not lead to a converging solution. But, if the eigenvalues of the Jacobi iteration matrix lie in an infinite strip

$$S = \{Z : |\text{Real}(Z)| < 1, |\text{Imag}(Z)| < \infty\}, \tag{2.3}$$

one can, following [2], develop a two-step iterative method. The idea is to construct an ellipse to envelope the eigenspectrum of the iteration matrix such that the minor axis is on the real interval $(-1,1)$ and the major axis is on the imaginary axis, and express the parameters associated with the two-step method in terms of M and m, the major and minor semiaxes, respectively.

In the usual way, let us split the matrix A as

$$A = D + E + F,$$

where D, E, and F are the diagonal, strictly lower triangular, and strictly upper triangular parts of A, respectively. $\tilde{A} = D^{-1}(E+F)$ is the usual Jacobi iteration matrix. Then the two-step method can be written as

$$U^{(k+2)} = \left(1 + \lambda\mu_1^2\right)\tilde{A}U^{(k+1)} - \lambda\mu_1^2 U^{(k)} + \left(1 + \lambda\mu_1^2\right)D^{-1}b, \tag{2.4}$$

where

$$\lambda = \frac{(m-M)}{(m+M)} \tag{2.4i}$$

and μ_1 is the unique root in $(0,1)$ of the quadratic equation

$$(M+m)\left(1+\lambda\mu^2\right) = 2\mu. \tag{2.4ii}$$

Our numerical study is carried out using (2.4) on the model problem (2.1) with a suitable $f(x,y)$ such that the solution to the problem on a unit square with zero boundary conditions is given by

$$U(x,y) = xe^{xy}\sin(\pi x)\sin(\pi y), \qquad (x,y) \in [0,1]\times[0,1]. \tag{2.5}$$

In the case $|Rh| > 2$, although the eigenspectrum of the Jacobi iteration matrix $\tilde{A}$ was seen to spread into the complex plane, the real parts of the eigenvalues were still confined in the interval $(-1,1)$. Hence, numerical computations with converging solutions were possible even for large $|R|$ values. Therefore, in order to employ the method (2.4), we calculate the values for the parameters λ and μ_1 using computed values for m and M, the minor and major semiaxes, respectively, of the ellipse enveloping the eigenspectrum of $\tilde{A}$. The method for obtaining the ellipse is explained in Section 4.

3. TWO-STEP ITERATION METHOD (GAUSS-SEIDEL TYPE) ON THE MODEL PROBLEM

We could reformulate the two-step method (2.4) in the following form:

$$(D+E)U^{(k+2)} = -\left(1+\lambda\mu_1^2\right)FU^{(k+1)} - (D+E)\lambda\mu_1^2U^{(k)} + \left(1+\lambda\mu_1^2\right)b \tag{3.1}$$

and carry out some numerical simulations on the model problem (2.1). In some cases, for large R values, the eigenspectra of the iteration matrix $-(D+E)^{-1}F$ will not lie in the infinite strip S. The symmetry about the imaginary axis is lost in these eigenspectra and the eigenvalues spread more into the left-hand complex plane with increasing R, though the real parts of the eigenvalues are always less than one.

If the ellipse enveloping the Gauss-Seidel iteration matrix has one of its axes along the real axis, and if $\tilde{m}$ and $\tilde{M}$ are the semiaxes along the horizontal and vertical directions, respectively, then a two-step method can be written as (see [3]),

$$(D+E)U^{(k+2)} = -\mu_0FU^{(k+1)} + (D+E)\mu_1U^{(k+1)} + (D+E)\mu_2U^{(k)} + \mu_0 b. \tag{3.2}$$

Here the parameters μ_0, μ_1, and μ_2 are given by

$$\mu_0 = \frac{2}{\tilde{m}+\tilde{M}}, \tag{3.2i}$$

$$\mu_1 = \frac{2\left(\tilde{m}-1\right)}{\tilde{m}+\tilde{M}}, \tag{3.2ii}$$

and

$$\mu_2 = \frac{\left(\tilde{M}-\tilde{m}\right)}{\tilde{m}+\tilde{M}}. \tag{3.2iii}$$

We calculate $\tilde{m}$ and $\tilde{M}$ in the following manner. Let

$$v_l = c_l + id_l, \qquad l = 1,\ldots,N \tag{3.3}$$

be the eigenvalues of the $N \times N$ iteration matrix $-(D+E)^{-1}F$. Now, if we define

$$c_{\max} = \max_l c_l,$$

then

$$\tilde{m} = 1 - \frac{1}{2}(c_{\max} - \rho) \tag{3.3i}$$

and $\tilde{M}$ is the solution of

$$\frac{(c_{\max} + \rho)^2}{4\tilde{m}^2} + \frac{\rho^2}{\tilde{M}^2} = 1 \tag{3.3ii}$$

(ρ is the spectral radius of the iteration matrix $-(D+E)^{-1}F$, and it can be estimated easily).

In the next section, we show how to construct the spectrum enveloping ellipse using the Arnoldi-Chebyshev Method.

4. THE ARNOLDI-CHEBYSHEV METHOD

The Arnoldi algorithm can be used to produce an upper Hessenberg matrix $H_m = V_m^\top \tilde{A} V_m$, where the columns of V_m, namely, $v_1, v_2, \ldots, v_m$ form an orthonormal basis [7]. The eigenvalues of the Hessenberg matrix will then provide approximations to the eigenvalues of $\tilde{A}$. This algorithm has the drawback of requiring the presence in memory of all previous vectors at a given step m and the amount of work increases tremendously with the step number m. In order to overcome this drawback, we use the Arnoldi algorithm iteratively. There are instances where the iterative Arnoldi algorithm performs poorly. In some cases, the minimum number of steps m that must be performed in each inner iteration in order to ensure convergence of the process is too large. In order to avoid these shortcomings of the iterative Arnoldi process, and to improve its overall performance, we use it in conjunction with the Chebyshev iteration (see [5]).

Let $\lambda_1, \ldots, \lambda_N$ be the eigenvalues of $\tilde{A}$ labeled in decreasing order of their real parts, and suppose that we are interested in λ_1.

The main part of this combined algorithm is the Chebyshev iteration, which computes a vector of the form $z_n = p_n(\tilde{A})z_0$, where n is the number of steps of the Chebyshev iteration, p_n is a polynomial of degree n, and z_0 is an initial vector. We would like to choose p_n in such a way that the vector z_n converges rapidly towards an eigenvector of $\tilde{A}$ associated with λ_1 as n tends to infinity. Therefore, we seek a minimal polynomial on the discrete set $P = \{\lambda_2, \lambda_3, \ldots, \lambda_N\}$ which satisfies the condition

$$p_n(\lambda_1) = 1.$$

Such a polynomial would be the one which minimizes the discrete uniform norm

$$\max_{\lambda \in P} |p(\lambda)|,$$

on the discrete set P over all polynomials of degree n satisfying the above condition. In order to compute such a polynomial, we need to know all the eigenvalues of $\tilde{A}$, which is unreasonable. Therefore, a more reasonable approach would be to replace the discrete minimax polynomial by the continuous one on a domain containing P, but excluding λ_1. Let E be such a domain in the complex plane, and let P_n denote the space of all polynomials of degree not exceeding n. Thus, the required polynomial p_n would be the one which achieves the minimum

$$\min_{p \in P_n,\ p(\lambda_1)=1} \ \max_{\lambda \in E} |p(\lambda)|.$$

To solve the above minimax problem explicitly, we restrict E to be an ellipse having its center on the real line, and containing the unwanted eigenvalues λ_i, $i = 2, \ldots, N$.

Let $E(d,c,a)$ be an ellipse with real center d, foci $d+c$, $d-c$, major semiaxis a or $\sqrt{a^2-c^2}$ depending on whether c is real or imaginary, and containing the set P. Since the spectrum of $\tilde{A}$ is symmetric with respect to the real axis, we will restrict $E(d,c,a)$ to be symmetric as well. In other words, the main axis of the ellipse must be either the real axis or must be parallel to the imaginary axis. Therefore, a is real and c is either real or purely imaginary. Then the best minimax polynomial is the polynomial

$$p_n(\lambda) = \frac{T_n[(\lambda-d)/c]}{T_n[(\lambda_1-d)/c]},$$

where T_n is the Chebyshev polynomial of degree n of the first kind.

The computation of z_n, $n=1,2,\ldots$ is simplified by the three-term recurrence for the Chebyshev polynomials

$$T_1(\lambda) = \lambda, \qquad T_0(\lambda) = 1,$$

and

$$T_{n+1}(\lambda) = 2\lambda T_n(\lambda) - T_{n-1}(\lambda), \qquad n = 1,2,\ldots.$$

Letting

$$\rho_n = T_n\left[\frac{(\lambda_1-d)}{c}\right], \qquad n = 0,1,\ldots,$$

we obtain

$$\rho_{n+1}p_{n+1}(\lambda) = T_{n+1}\left[\frac{(\lambda-d)}{c}\right] = 2\frac{\lambda-d}{c}\rho_n p_n(\lambda) - \rho_{n-1}p_{n-1}(\lambda).$$

Let us transform this further by setting $\sigma_{n+1} = \rho_n/\rho_{n+1}$,

$$p_{n+1}(\lambda) = 2\sigma_{n+1}\frac{\lambda-d}{c}p_n(\lambda) - \sigma_n\sigma_{n+1}p_{n-1}(\lambda).$$

A straightforward manipulation using the definitions of σ_i, ρ_i and the three-term recurrence relation of the Chebyshev polynomials shows that σ_i, $i=1,\ldots$, can be obtained from the recursion

$$\sigma_1 = \frac{c}{(\lambda_1-d)}, \qquad \sigma_{n+1} = \frac{1}{2/\sigma_1-\sigma_n}, \qquad n = 1,2,\ldots.$$

The above two recursions can now be assembled together for computing

$$z_i = p_i\left(\tilde{A}\right)z_0, \qquad i = 1,2,\ldots,$$

i.e.,

$$z_{n+1} = 2\frac{\sigma_{n+1}}{c}\left(\tilde{A}-dI\right)z_n - \sigma_n\sigma_{n+1}z_{n-1}.$$

Although λ_1 is not known, we can replace it by some approximation ν in practice.

The polynomial p_n amplifies the components of z_0 in the direction of the desired eigenvectors while damping those in the remaining eigenvectors. Once $z_n = p_n(\tilde{A})z_0$ is computed, a few steps of Arnoldi's method, with $v_1 = z_n/\|z_n\|$ as the initial vector are carried out in order to extract from z_n, the desired eigenvalues.

Suppose that we can find an ellipse $E(d,c,a)$ with real center d, foci $d+c$, $d-c$, and major semiaxis a or $\sqrt{a^2-c^2}$, depending on whether c is real or imaginary, that contains all the eigenvalues of $\tilde{A}$ except the r eigenvalues of $\tilde{A}$ with largest real parts. We will describe later a procedure for obtaining such an ellipse [8]. Then, we run a certain number of steps of the Chebyshev iteration and take the resulting vector z_n as initial vector in the Arnoldi process. From the Arnoldi purification process, one obtains a set of m eigenvalues, r of which are approximations to the r wanted ones, while the remaining ones will be useful for adaptively constructing the best ellipse. After a cycle consisting of n steps of the Chebyshev iteration followed by m steps of

the purification process, if the accuracy realized for the r rightmost eigenpairs is not sufficient, restarting will then be necessary. The following is an outline of a simple algorithm based on the above ideas.

- Step 1: Choose an arbitrary initial vector v_1; m, the number of Arnoldi steps and n, the number of Chebyshev steps.
- Step 2: Perform m steps of the Arnoldi algorithm starting with v_1. Compute the m eigenvalues of the resulting Hessenberg matrix. Select the r eigenvalues with largest real parts $\tilde{\lambda}_1, \ldots, \tilde{\lambda}_r$ and take $\tilde{P} = \{\tilde{\lambda}_{r+1}, \ldots, \tilde{\lambda}_m\}$. If sufficient accuracy is achieved stop, otherwise continue.
- Step 3: Using $\tilde{P}$, obtain the new estimates of the parameters d and c of the best ellipse. Then compute the initial vector z_0 for the Chebyshev iteration as a linear combination of the approximate eigenvectors.
- Step 4: Perform n steps of the Chebyshev iteration to obtain z_n. Take $v_1 = z_n/\|z_n\|$ and go back to Step 2.

4.1. Obtaining the Best Ellipse

We would like to find the optimal ellipse, the one with smallest convergence factor

$$\tau\left(d, c^2\right) = \frac{a + \left(a^2 - c^2\right)^{1/2}}{d + \left(d^2 - c^2\right)^{1/2}} \tag{4.1}$$

(see [9]), enclosing the set P of the eigenvalues other than the ones with the r largest real parts, i.e., we need to find the optimal parameters d and c. We use the procedure due to Manteuffel [9] to find such d and c.

Let $S^+ = \{\lambda_i \in S \mid \mathrm{Im}(\lambda_i \geq 0\}$. We will refer to S^+ as the *positive hull* of the spectrum.

Now, suppose the positive hull contains two eigenvalues $\lambda_1 = x_1 + iy_1$, $\lambda_2 = x_2 + iy_2$. Then the mini-max solution must occur when the eigenvalues λ_1 and λ_2 lie on the same member of the family $E(d, c, a)$ in the λ-plane. They both must satisfy the equation of the ellipse

$$\frac{(d-x)^2}{a^2} + \frac{y^2}{a^2 - c^2} = 1.$$

Hence, the following two equations can be solved simultaneously to obtain d and c:

$$\frac{(d-x_1)^2}{a^2} + \frac{y_1^2}{a^2 - c^2} = 1$$

and

$$\frac{(d-x_2)^2}{a^2} + \frac{y_2^2}{a^2 - c^2} = 1.$$

Let

$$C = \frac{x_2 - x_1}{2}, \qquad B = \frac{x_2 + x_1}{2},$$
$$S = \frac{y_2 - y_1}{2}, \qquad T = \frac{y_2 + y_1}{2}.$$

We may assume that $x_2 > x_1$. Then, we have $C > 0$, $B > 0$, $T \geq 0$, and the solutions are given by the following relationships.

If $S = 0$, then

$$d = B \quad \text{and} \quad c^2 = \frac{a^2\left(a^2 - \left(C^2 + T^2\right)\right)}{\left(a^2 - C^2\right)}, \tag{4.2}$$

where $y = a^2$ can be found as the *only* real root of the cubic polynomial

$$q_1 y^3 + q_2 y^2 + q_3 y + q_4 = 0$$

in the interval (C^2, B^2). The coeffecients are

$$\begin{aligned} q_1 &= B^2 + T^2, \\ q_2 &= -3C^2B^2, \\ q_3 &= 3C^4B^2, \\ q_4 &= -C^4B^2\left(C^2 + T^2\right). \end{aligned}$$

If $S \neq 0$, then

$$c^2 = \frac{(d - (B + ST/C))(d - (B - (CT/S))(d - BCS/T)}{(d - B)}$$

and

$$a^2 = (d - (B - CT/S))(D - (B - CS/T)),$$

where d can be found in terms of $z = d - B$, as the root of the polynomial

$$p_1 z^5 + p_2 z^4 + p_3 z^3 + p_4 z^2 + p_5 z + p_6 = 0,$$

in the interval $(0, C)$ for $S > 0$ and $(-C, 0)$ for $S < 0$.

The coeffecients are

$$\begin{aligned} p_1 &= \left(2B - C\left(\frac{T}{S} + \frac{S}{T}\right)\right)\left(2B + \frac{ST}{C} - C\left(\frac{T}{S} + \frac{S}{T}\right)\right), \\ p_2 &= \left(2B + \frac{ST}{C} - C\left(\frac{T}{S} + \frac{S}{T}\right)\right)\left((2BC + ST)\left(\frac{T}{S} + \frac{S}{T}\right) + 4C^2\right) \\ &\quad + B^2\left(2B - C\left(\frac{T}{S} + \frac{S}{T}\right)\right) + B\left(B^2 - C^2\right), \\ p_3 &= 4C^4 - 4C^3B\left(\frac{T}{S} + \frac{S}{T}\right) + C^2ST\left(\left(\frac{T^3}{S^3} + \frac{S^3}{T^3}\right) - 3\left(\frac{T}{S} + \frac{S}{T}\right)\right) \\ &\quad + C^2B^2\left(\frac{T^2}{S^2} + \frac{S^2}{T^2} + 3\right), \\ p_4 &= CST\left(\left(B - C\frac{T}{S}\right)\left(B - 3C\frac{T}{S}\right) + \left(B - C\frac{S}{T}\right)\left(B - 3C\frac{S}{T}\right)\right), \\ p_5 &= -3C^3ST\left(2B - C\left(\frac{T}{S} + \frac{S}{T}\right)\right), \\ p_6 &= -3C^3S\left(B^2 - C^2\right). \end{aligned}$$

The point thus found represents the best parameter values when the positive hull contains two eigenvalues. Such a point will be referred to as a *pair-wise best point*, and the associated ellipse passing through the two eigenvalues will be referred to as the *pair-wise best ellipse*. For each pair of eigenvalues in the positive hull, we find the pair-wise best point. The pair-wise best point with the smallest convergence factor (4.1) and whose ellipse contains the other members of the positive hull in the closure of its interior, is the mini-max solution.

4.2. Selection of the Initial Vector z_0 for the Chebyshev Iteration

Let us now indicate how one can select the initial vector z_0. It is desirable to start the Chebyshev iteration by a vector which is a linear combination of the approximate eigenvectors associated with the rightmost r eigenvalues.

Let ξ_i $(i = 1, \ldots, r)$ be the coefficients of the desired linear combination. Then the initial vector for the Chebyshev process is

$$z_0 = \sum_{i=1}^{r} \xi_i \tilde{u}_i = \sum_{i=1}^{r} \xi_i V_m \tilde{y}_i = V_m \sum_{i=1}^{r} \xi_i \tilde{y}_i. \tag{4.3}$$

Hence,

$$z_0 = V_m y, \qquad \text{where } y = \sum_{i=1}^{r} \xi_i \tilde{y}_i.$$

$\tilde{u}_i = V_m \tilde{y}_i$ are the approximate eigenvectors of $\tilde{A}$ and $\tilde{y}_i$ is an eigenvector of H_m associated with the eigenvalue $\tilde{\lambda}_i$. We need to only compute the $\tilde{y}_i$s and select the appropriate ξ_is. We choose $\xi_i = \|(\tilde{A} - \tilde{\lambda}_i I)\tilde{u}_i\|$. Notice that the ξs corresponding to two complex conjugate approximate eigenelements are equal, so that this choice of ξs will lead to real z_0.

4.3. Choosing the Parameters m and n

The number of Arnoldi steps m and the number of Chebyshev steps n are important parameters that affect the effectiveness of the method. Since we want to obtain more eigenvalues than the r desired ones, in order to use the remainder in choosing the parameters of the ellipse, m should be at least $r + 2$. In fact, it would be preferable to have m several times larger than r at least three or four times.

When choosing n, we have to make sure that it is neither too small nor too large when compared to the dimension of the iteration matrix. For if n is too small, it may result in slowing down the algorithm. If n is too large, the vector z_n may become nearly an eigenvector which could be troublesome for the Arnoldi process. Moreover, the parameters d and c of the ellipse may be far from optimal and one will have to re-evaluate them frequently.

5. APPLICATION TO THE MODEL PROBLEM

In this section, we present the findings based on the numerical computations carried out on the model problem (2.1) making use of the two-step iteration methods (2.4) and (3.2), in conjunction with the Arnoldi-Chebyshev algorithm.

Figures 1 and 2 give the eigenspectra of the Gauss-Seidel iteration matrix and the corresponding ellipses when R takes the values 35 and 57, respectively, and when $\alpha = 0.5$. Note that, when $\alpha = 0.5$, the convection term will have a central difference discretization. A number of computations were performed for $\alpha = 0.5$ using the two-step Gauss-Seidel type method (3.2) on the model problem (2.1) with the $f(x, y)$ corresponding to (2.5). We were able to obtain the coverged solution as competitively or faster than compared to the method by Manoranjan and Drake [3], where 0.5 was used as an estimate for $c_{\max}$ (see Table 1).

Table 2 gives the computations for $R = 200$ and for different values of α. Even though the error is the least when $\alpha = 0.43$, it would be desirable to use either $\alpha = 0.25$ or 0.3, since the convergence is much faster with a reasonable error.

Computations were performed for different values of the upwind parameter α again using (3.2) on the model problem (2.1) with the same $f(x, y)$. As expected, for large R values, we are able to obtain the converged solution as competitively or faster than compared to the two-step Jacobi type method (2.4). Table 3 gives the comparison between the two methods. Optimal α was taken to be the α value which gave the least number of iterations.

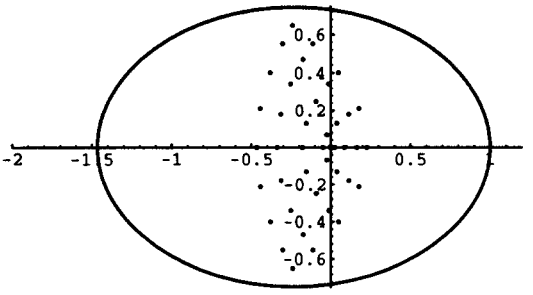

Figure 1. Eigenspectrum of $-(D+E)^{-1}F$ when $\alpha = .5$ and $R = 35$.

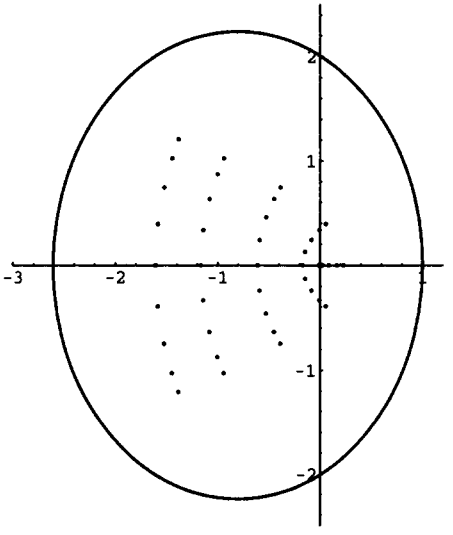

Figure 2. Eigenspectrum of $-(D+E)^{-1}F$ when $\alpha = .5$ and $R = 57$.

Table 1.

Rh	Number of Iterations	
	$c_{\max} = 0.5$	Computed $c_{\max}$
4.0	34	55
5.0	28	25
6.0	49	28
8.0	109	61
10.0	243	149
12.0	465	278

Table 2.

α	No. of Iterations	Relative Error (l_2 norm)
0.00	73	0.1668
0.10	127	0.1356
0.15	43	0.1196
0.20	25	0.1033
0.25	16	0.0871
0.30	27	0.0710
0.35	58	0.0561
0.40	114	0.0446
0.43	222	0.0417
0.45	311	0.0429

Table 3.

R	Jacobi Method			Gauss-Seidel Method		
	Optimal α	Number of Iterations	Relative Error (l_2 norm)	Optimal α	Number of Iterations	Relative Error (l_2 norm)
200	0.06	16	0.1482	0.26	16	0.0838
300	0.05	15	0.1526	0.24	15	0.0914
400	0.03	15	0.1595	0.22	14	0.0985
500	0.03	15	0.1599	0.21	14	0.1021
700	0.03	15	0.1604	0.21	14	0.1025
1000	0.03	15	0.1607	0.20	13	0.1061
2000	0.02	14	0.1642	0.19	12	0.1098
5000	0.02	14	0.1645	0.19	13	0.1100

In Figure 3, we present the eigenspectrum and the corresponding ellipse when $R = 200$ and $\alpha = 0.25$. Here $\rho = 1.19$, and it seems that the ellipse is an overestimate and one might think that the circle with radius ρ would give us a better solution. In fact, the ellipse obtained seems to be an overestimate than the circle with radius ρ for large R values. But as can be seen from Table 4, for several large R values faster convergence was obtained with the ellipse rather than the circle with radius ρ.

In all the computations, h is chosen as 0.1 and the initial guess is the zero vector.

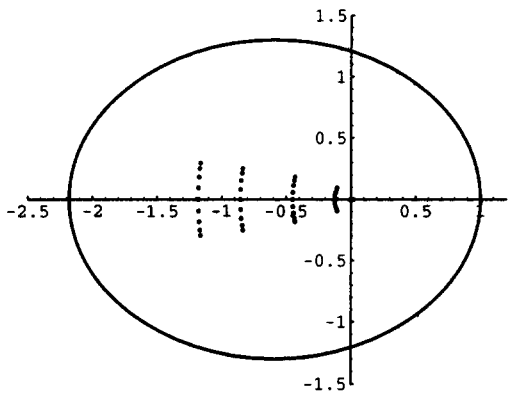

Figure 3. Eigenspectrum of $-(D+E)^{-1}F$ when $\alpha = .25$ and $R = 200$.

Table 4.

R	Number of Iterations	
	Circle with Radius ρ	Ellipse with Semiaxes $\tilde{m}$ and $\tilde{M}$
200	22	16
300	22	15
400	22	14
500	22	14
700	21	14
1000	21	13
2000	21	12
5000	21	13

6. CONCLUSION

Overall, our findings indicate that using the $c_{\max}$ obtained from the Arnoldi-Chebyshev method is better than using the estimate 0.5 in the Gauss-Seidel two-step method for the model convection diffusion problem. We have also demonstrated that discretizing the convection term by an

upwind formula gives us a much faster converging solution than central differencing, in addition to avoiding any unwanted numerical oscillations. So, in this paper, we have improved the iteration method presented in [3] by providing a systematic way of constructing the ellipse enveloping the eigenspectrum of the corresponding iteration matrix. As demonstrated by our numerical computations, the systematic two-step iteration method described here can be a useful and cost-effective method for solving a system of linear equations, especially when the standard iteration methods fail.

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